

Set -1

HDFS (Storage)

A. Hadoop Storage File system

I. Write a command to create the directory structure in HDFS.

II. Write a Command to move files from the local Unix/Linux machine to HDFS.

B. Viewing Data Contents, Files and Directory

a) Write an HDFS command to see the contents of files in HDFS.

b) Write an HDFS command to see the contents of files which are present in HDFS.

C. Getting Files data from the HDFS to Local Disk

I. Write an HDFS command to copy the file from HDFS to the local file system.

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Map Reduce Programming (Processing data).

A. Word Count

Develop a MapReduce example program in a MapReduce environment to find out the number of occurrences of each word in a text file

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Data Processing Tool – Hive (NOSQL query-based language)

We have a table 'user data' that contains the following fields:

data date: string

user_id: string

properties: string

The properties field is formatted as a series of attribute=value pairs.

Ex: Age=21; state=CA; gender=M;

i. Create the table in HIVE using hive no sql based query.

ii. Fill the table with sample data by using some sample databases.

iii. Write a program that produces a list of properties with minimum value(min_value), largest value(max_value) and number of unique values. Before you start, execute the prepare step to load the data into HDFS.

iv. Generate a count per state.

Write a program that lists the states and their count from the data input.

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Data Processing Tool – Pig (Latin based scripting lang)

A. Simple Logs

We have a set of log files and need to create a job that runs every hour and perform some calculations.

The log files are delimited by a 'tab' character and have the following fields:

- a) site
- b) hour_of_day
- c) page_views
- d) data_date

The log files are located in the prepare folder. Load them in HDFS at data/ pig/simple logs folder and use them as the input.

Important: In order to load tab delimited files use pigStorage ('\u0001').

Create a program to:

- i. Calculate the total views per hour per day.
- ii. Calculate the total views per day.
- iii. Calculate the total counts of each hour across all days.
- iv. We can write word count script by passing a text file as input.

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SQOOP

It is used to import and export data from SQL to HDFS and Vice versa. Before processing any data first need to import data into HDFS. As of now seen importing files into HDFS but using sqoop tool we can import SQL table data into hive table.

- I. Create table in HIVE using hive query language.
- II. Import the sql table data into hive using SQOOP tool.
- III. Export hive table data into local machine and into SQL.

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Exploring the ARIMA model.